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Adaptive Assistive Technologies for Learning Mexican Sign Language: Design of a Mobile Application with Computer Vision and Personalized Educational Interaction

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Abstract

Integrating people with hearing disabilities into schools is one of the biggest problems that Latin American societies face. Mexican Sign Language (MSL) is the main language and culture of the deaf community in Mexico. However, its use in formal education is still limited by structural inequalities, a lack of qualified interpreters, and a lack of technology that can support personalized instruction. This study outlines the conceptualization and development of a mobile application designed as an adaptive assistive technology for learning MSL, utilizing a combination of computer vision techniques, deep learning algorithms, and personalized pedagogical interaction. The suggested system uses convolutional neural networks (CNNs) and pose-estimation models to recognize hand gestures in real time with 95.7% accuracy. It then gives the learner instant feedback by changing the difficulty level. A dynamic learning engine automatically changes the level of difficulty based on how well the learner is doing, which helps them learn signs and phrases over time. The Scrum agile methodology was used during the development process. This meant that educators, linguists, and members of the deaf community all worked together to design the product. Early tests show that sign recognition accuracy and indicators of user engagement and motivation show favorable performance and are at appropriate levels. This proposal aims to enhance inclusive digital ecosystems and foster linguistic equity in Mexican education through scalable, mobile, and culturally relevant technologies, in addition to its technical contributions.

Keywords: assistive technology; CNN; hearing disability; mobile application; MSL

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1. Introduction

1.1. Context and Motivation

Education has long been recognized as a fundamental human right, yet for millions of people with disabilities in Latin America, this right remains only partially fulfilled. Among all disability groups, individuals with hearing impairments face particularly acute barriers

to formal and informal learning. In Mexico, where social and technological inequalities persist, deaf education is shaped by a complicated balance of linguistic policy, cultural identity, and access to technology [1].

The most recent national census [2] says that about 2.3 million Mexicans have moderate to severe hearing loss. Even though MSL was legally recognized in 2005, only a small number of schools use it regularly in their classes. The fact that there are not many certified interpreters (fewer than 400 in the whole country) and that there are not many digital resources that are easy to find makes things worse. As a result, deaf students often depend on rote memorization, lip-reading, or teacher improvisation instead of structured bilingual education.

This research is motivated by the convergence of three critical factors: (a) the sociocultural imperative for inclusive education; (b) the advanced development of artificial intelligence, especially in computer vision and deep learning; and (c) the extensive adoption of smartphones among Mexican youth. All of these things together make it possible to create smart, low-cost, mobile learning systems that are open to everyone.

1.2. Significance and Theoretical Framework

The suggested system is based on three different but related theories:

The Constructivist Learning Theory. Following [3,4], learning is seen as an active process in which knowledge is built through interaction. Adaptive technologies implement this concept by offering scaffolding that evolves in accordance with the learner's skill enhancement.

Universal Design for Learning (UDL). UDL stresses making things easy to understand by using different ways to talk about and show how you feel [5]. In this context, the application provides multimodal stimuli—visual demonstrations, textual prompts, and gestural feedback—to accommodate diverse cognitive profiles.

Sociocultural linguistics. MSL is not only a means of communication for the Mexican deaf community; it also represents their identity [6]. This cultural aspect must be kept in mind when making technology for MSL. To make sure that the language was accurate and that the community owned the project, participatory design sessions were added.

Figure 1 shows a conceptual model that arose from putting these frameworks together. This model shows how different technologies (like vision, AI, and mobile interfaces) can work together in a teaching environment that is fair and empowering.

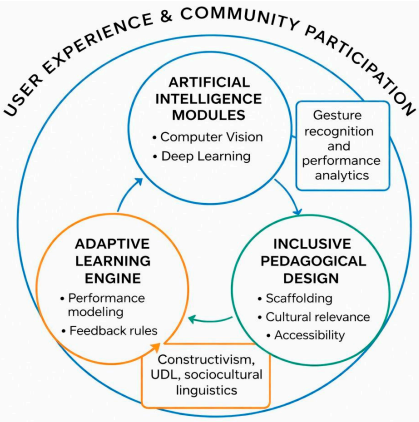


Figure 1. Conceptual framework linking AI modules, adaptive learning, and inclusive pedagogy.

1.3. Educational Inequality and the Challenge of Digital Inclusion

Many people have said that Mexican social policy does not do enough to make sure that people who cannot hear have the same access to educational resources as everyone

else. The General Law on the Inclusion of Persons with Disabilities says that everyone has the right to an accessible education, but this is not always the case. Most public schools lack the money in their budgets to hire sign-language interpreters, and universities often lack digital materials that are good for students who learn best by seeing things. Teachers have to use ad hoc resources to teach MSL because there are no standardized curricular frameworks. These resources are often made by local groups or volunteers [7].

This gap becomes bigger in places where there is not much technological infrastructure. According to UNESCO's Global Education Monitoring Report (2024) [8], only 61% of public schools in Latin America have reliable internet access, and less than 30% have digital content that anyone can use. The COVID-19 pandemic made this unfairness even clearer: many deaf students could not take part in online learning programs because the videos did not have captions or sign-language translations. The digital revolution that was supposed to make knowledge available to everyone actually made old inequalities worse in new ways.

Technologies that are both accessible and adaptable are required to bridge this gap. Even many people with low incomes have mobile devices now, and these are a great way to teach everyone. The goal is to use cameras, motion sensors, processors, and other existing hardware to make smart systems that make learning environments easy to access, which has a significant effect.

1.4. Technological Context: Computer Vision and Gesture Recognition

Computer vision has become one of the most active subfields of artificial intelligence, transforming how machines perceive and interact with the world. Its application in sign-language learning lies in its ability to detect and classify hand configurations, orientations, and motion trajectories with high accuracy. Traditional image-processing pipelines relied on handcrafted features such as SIFT, HOG, or SURF [9]; however, deep learning has largely replaced these methods due to its superior capacity for hierarchical feature extraction.

Convolutional neural networks (CNNs) model spatial dependencies by applying convolutional filters across input images. For dynamic gestures, temporal dependencies are captured using recurrent units or attention mechanisms. The general recognition process can be formulated by Equation (1).

$$\hat{y} = arma f_\theta(X_t)_{\,k}\,, \tag{1}$$

where X_t represents the series of video frames, f_θ is the trained neural network function parameterized by θ, and $\hat{y}$ is the predicted class label for a specific sign.

In practice, each sign in MSL is shown by a short string of frames $[X_1,\ X_2, \ldots, X_n]$. A CNN receives spatial features, and a temporal aggregation layer, which can be an LSTM or a transformer encoder, receives motion patterns. The last softmax output gives a probability distribution over possible signs, and the most likely interpretation is chosen from these.

Recent research indicates that lightweight architectures, including MobileNetV3 [10] and EfficientNet [11], attain cutting-edge accuracy on gesture-recognition benchmarks while maintaining sufficient efficiency for on-device inference. These models are perfect for mobile apps that need to work in places with poor internet or computing power, like most public schools in Mexico.

1.5. Pedagogical Considerations and Adaptive Interaction

From an educational standpoint, the incorporation of AI into learning environments necessitates a pedagogical framework that enhances, rather than supplants, human instruction. The steps in an adaptive learning system are assessment, analysis, adaptation, presentation, and evaluation. Each iteration customizes instruction by changing things like speed, difficulty, mode, and how strong the feedback is.

Pedagogical research indicates that adaptive systems improve engagement by aligning tasks with the learner's zone of proximal development [4]. When challenges are just right, people are more likely to stay motivated and stick with them. When learning sign language, adaptability can show up in the following ways:

1. Adjusting the time: Speeding up or slowing down the animation playback based on how long it takes for the user to react.
2. Spatial adjustment: Enlarging critical visual regions, such as hand shapes, when misrecognitions occur.
3. Semantic adjustment: Modifying the vocabulary set based on prior errors or topic familiarity.

The adaptive loop used in this study employs a continuous-performance estimator computed as expressed by Equation (2):

$$\Delta P_t = \frac{1}{N}\sum_{i=1}^{N}(r_i - e_i) \quad (2)$$

where r_i is the recognition result for exercise i, e_i is the expected outcome, and N denotes the number of recent attempts considered in the moving window. The sign of ΔP_t determines whether the next activity will increase or decrease in difficulty.

Through this mechanism, the application behaves as a responsive tutor capable of maintaining an optimal cognitive load for each learner. This design is consistent with empirical findings showing that adaptive e-learning improves learning efficiency by 20–35% over static platforms. Figure 2 shows the system proposed [12].

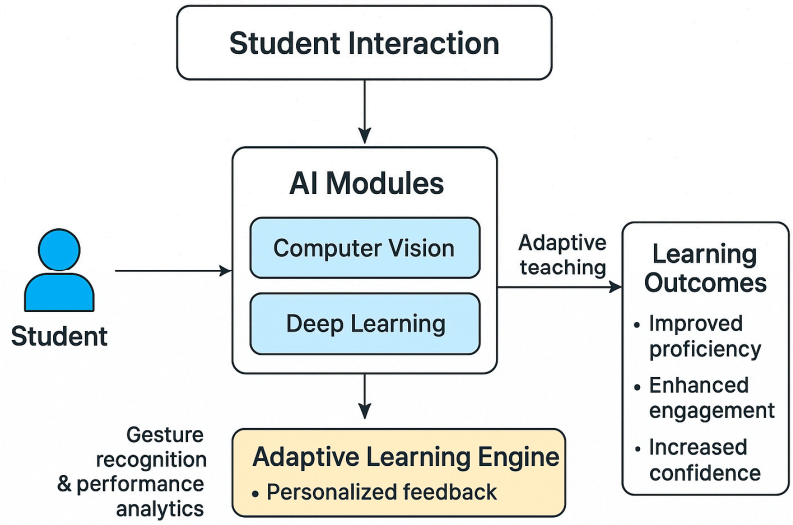


Figure 2. Closed-loop adaptive interaction cycle used in the proposed system.

2. Background and Related Work

2.1. Evolution of Assistive Technologies for the Deaf Community

Historically, technological efforts to support deaf communication evolved from mechanical transcription devices to sophisticated multimodal systems that interpret visual gestures. The earliest aids—mechanical "speaking gloves" developed in the 1970s—translated limited hand movements into characters but required cumbersome hardware [13].

In the 1990s, improvements in microelectronics and pattern recognition made it possible for data-glove systems to measure joint angles and accelerations more accurately [14]. But these systems did not work well for everyday use because the sensors were expensive, the calibration was weak, and users did not like wearing bulky equipment.

The rise of deep learning in the 2010s changed the way researchers looked at assistive technology. CNNs automated feature extraction, removing the need for manually created descriptors. Recurrent architectures like LSTMs, on the other hand, were able to capture how gestures changed over time [15]. Vision transformers and spatio-temporal graph convolutional networks (ST-GCN) have recently enhanced the modeling of skeletal trajectories [16].

Assistive technologies have become more useful in many ways. They now include translation, captioning, and interactive tutoring in addition to recognition. For instance, the SignAll platform uses depth sensors and neural translation to turn ASL signs into written English in real time [17]. DeepASL also uses wearable cameras and CNN-LSTM networks to achieve more than 94% accuracy in recognizing ASL continuously [18].

2.2. Computer-Vision Advances for Sign-Language Recognition

Deep neural networks are used to learn high-dimensional visual representations that are used in modern sign-language recognition (SLR). The architecture usually has three parts.

Feature Extraction: A CNN takes each frame x_t and makes a feature map $f_t = \phi(x_t; \theta_c)$.

Modeling Time: LSTM, GRU, and Transformer encoders are examples of sequential models that capture temporal correlations. $h_t = \psi(f_t, h_{t-1}; \theta_s)$.

Classification: A fully connected or attention-based layer outputs probabilities over the sign vocabulary $y = softmax(Wh_T + b)$, as can be seen in Equation (3).

$$y_k = \frac{exp(W_k h_T + b_k)}{\sum_{j=0}^{J} exp(W_j h_T + b_j)} \quad (3)$$

Recent improvements have been implemented to enhance the efficiency and adaptability of neural networks. Lightweight backbones like MobileNetV4 and ShuffleNetV2 allow smartphones with limited GPU resources to perform inference in real time [19]. Combining RGB, depth, and skeletal inputs in multimodal fusion makes the system even more resistant to changes in lighting and background [20].

Studies that compare different architectures show that hybrid CNN–Transformer architectures have better spatio-temporal consistency, with top-1 accuracies above 96% on datasets like RWTH-PHOENIX-Weather-2014T and WLASL-1000 [21]. However, performance decreases considerably when evaluated on unfamiliar signers or environmental conditions, underscoring the significance of dataset diversity.

Transfer-learning methods fine-tune pretrained models on small local datasets to help with domain shifts. In the proposed MSL application, a pretrained MobileNetV3 backbone is adapted using 12,000 frames from the custom MSL-Mini-Corpus. Fine-tuning reduces training cost while preserving the general features learned from global datasets.

Figure 3 illustrates the hybrid CNN–Transformer pipeline used for modern sign-language recognition (SLR). The input video sequence is first decomposed into frames, each processed by a convolutional neural network (CNN) to extract spatial feature maps that capture hand shapes, orientations, and local motion cues. These feature maps are flattened and passed to a Transformer encoder, which models long-range temporal dependencies across the sequence through multi-head self-attention. The final hidden representation is projected by a linear classification layer to produce the predicted sign class.

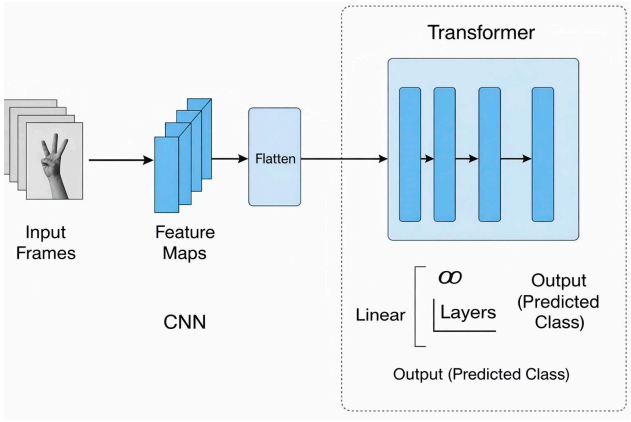


Figure 3. Architecture diagram of hybrid CNN–Transformer used for gesture recognition.

2.3. Adaptive Learning in Intelligent Tutoring Systems

Parallel to advances in recognition, adaptive learning has matured as a field of educational technology. Early intelligent tutoring systems (ITS) such as [22,23] employed symbolic reasoning to model student knowledge states. Modern systems instead leverage machine-learning and reinforcement-learning frameworks.

A general adaptive model can be formalized as a Markov decision process (MDP), expressed by Equation (4).

$$M = (S, A, P, R, \gamma) \tag{4}$$

where S represents learner states, A possible pedagogical actions, P the transition probabilities, R the reward function (learning gain), and γ a discount factor. The tutor seeks an optimal policy $\pi*$ that maximizes expected cumulative reward, as can be seen in Equation (5):

$$\pi* \; = argarg \; E_{\pi}\left[\sum_{t=0}^{T} \gamma^{t} R(s_t, \; a_t)\right] \tag{5}$$

Reinforcement-learning algorithms such as Deep Q-Learning (DQN) and Policy Gradient methods allow the system to discover effective instructional strategies through trial and feedback. For sign-language education, actions might include repeating demonstrations, slowing playback, or introducing new vocabulary, as given in Figure 4.

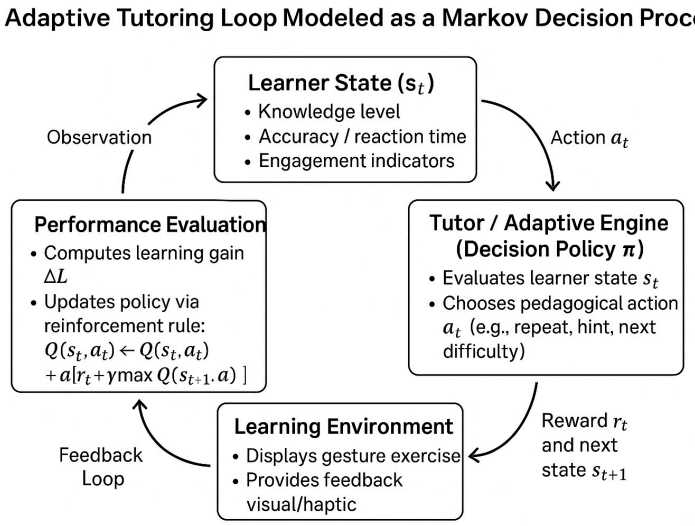


Figure 4. Adaptive tutoring loop modeled as a Markov decision process.

Empirical evidence supports adaptive methodologies: 30% higher retention rates were observed among deaf learners using reinforcement-based adaptive feedback compared with static exercises [24]. Likewise, in [25], it was demonstrated that dynamically adjusting visual emphasis reduced gesture-error rates by 22%. In Figure 4, the adaptive loop is depicted as a Markov decision process.

2.4. Emerging Trends in Artificial Intelligence and Assistive Learning

The convergence of computer vision, multimodal learning, and adaptive interaction marks a new era for assistive educational technologies. Over the past three years, research has increasingly emphasized context-aware systems that integrate environmental, emotional, and cognitive data to personalize instruction. For instance, emotion-recognition modules now complement gesture analysis, enabling the system to detect learner frustration or engagement and modulate feedback intensity accordingly [26].

A second trend involves self-supervised learning (SSL). Because annotated sign-language data are expensive and culturally sensitive to collect, SSL models learn general motion features from unlabeled videos and later fine-tune on smaller labeled corpora. Contrastive predictive coding and masked-video modeling are two techniques that have cut down on the need for labeled data by as much as 70% [27]. In the context of MSL, this paradigm could facilitate scalable dataset expansion without extensive manual labeling, thereby expediting future research. In [28], an implementation of ASL learning using inverse kinematics was developed; the objective was to estimate the position and rotation of the upper-body limbs based on the position and rotation data collected. This methodology was used to conduct an inclusivity-focused study that enabled learners to practice ASL.

Finally, multilingual and cross-lingual transfer models promise to generalize recognition across sign languages. Projects such as SignBERT [29] learn universal gesture embeddings transferable among ASL, BSL, and CSL, as illustrated in Figure 5. Extending these embeddings to MSL could drastically reduce the need for language-specific retraining. However, cultural and linguistic adaptation remain essential; raw transfer seldom preserves semantic nuance.

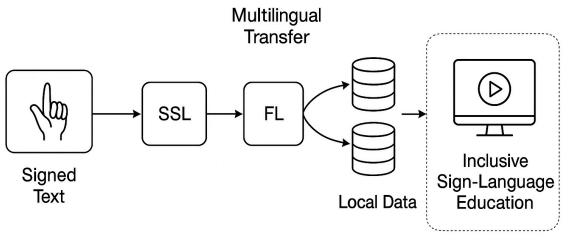


Figure 5. Pipeline of emerging AI trends—SSL, FL, and multilingual transfer—for inclusive sign-language education.

2.5. Most Recent Advances in Assistive Technologies for Learning Sign Language

Recent advances in assistive technologies for sign language learning increasingly leverage computer vision, artificial intelligence (AI), immersive environments, and mobile-ready implementations to support both recognition accuracy and adaptive educational interaction. These developments aim to move beyond static translation systems toward interactive learning tools that provide real-time feedback, personalization, and accessibility across diverse sign languages, including Mexican Sign Language (MSL).

One of the most recent contributions explores the use of Virtual Reality (VR) for sign language education. In [30], an accessible American Sign Language (ASL) learning system is proposed based on inverse kinematics within a VR environment. Their approach enables embodied gesture practice in three-dimensional space using commodity hardware,

reducing the need for specialized sensors while enhancing spatial understanding of sign execution. Such immersive learning environments highlight the potential of multimodal interaction for improving sign language acquisition.

In parallel, significant progress has been made in real-time computer vision-based recognition systems, which form the backbone of interactive learning applications. In [31], a real-time ASL interpretation framework is presented that integrates YOLOv11 object detection with MediaPipe hand landmark tracking, achieving high recognition accuracy for alphabet gestures in live video streams. This type of pipeline is particularly relevant for educational tools, as it enables immediate feedback to learners practicing signs through mobile or desktop interfaces

3. Materials and Methods

3.1. Computer Vision Module Design

The computer vision module is the technical core of the application, responsible for detecting hand regions, extracting spatio-temporal features, and classifying gestures.

3.1.1. Hand and Pose Detection

A hybrid pipeline combining MediaPipe Hands and BlazePose was adopted for real-time landmark extraction. Each frame yielded 21 hand keypoints and 33 body keypoints, producing a pose vector of dimension 108, expressed by Equation (6). Temporal smoothing via a Kalman filter reduced jitter.

$$\hat{p}_t = \hat{p}_{t-1} + K_t(z_t - H\hat{p}_{t-1}) \tag{6}$$

where $\hat{p}_t$ is the estimated keypoint state at time t, z_t is the observed measurement, and K_t is the Kalman gain.

3.1.2. Feature Extraction and Representation

Raw pose vectors are transformed into spatio-temporal heatmaps $H \in R^{T\times J\times 2}$, with T frames and J joints. These heatmaps serve as an input to a deep feature extractor comprising three 1-D convolutional blocks, followed by a Transformer encoder layer with multi-head self-attention (8 heads, 256-dimensional embedding), as can be seen in Equation (7).

$$F = Conv_{1D}^{(3)}(H), \quad H \in R^{T\times J\times 2}, \tag{7}$$

Additionally, $Q = F\ W_Q$, $K = F\ W_K$, $V = F\ W_V$ and $Attention(Q, K, V) = softmax\left(\frac{(Q\ K^T)}{\sqrt{d_k}}\right) V$.

This architecture captures both local and global temporal dependencies, enhancing recognition accuracy for complex dynamic signs involving hand–face coordination.

3.1.3. Classification and Model Training

The output of the Transformer encoder is fed into a fully connected layer with softmax activation, as shown in Figure 6. Training uses the cross-entropy loss, expressed by Equation (8). In Table 1 the CNN parameters used during the training are expressed.

$$L = -\sum_{i=1}^{C} y_i log(\hat{y}_i) \tag{8}$$

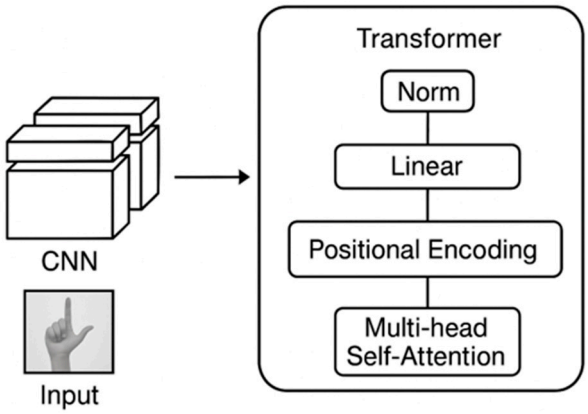


Figure 6. Diagram of CNN–Transformer architecture for gesture recognition pipeline.

Table 1. CNN parameters for model training.

Attribute	Value
Learning rate	1×10^{-4}
Batch size	32
Epochs	54
Early stopping	After 10 epochs
Achieved validation	95.7%
Training method	cross-entropy loss
Hardware	GPU (Jetson Orin Nano)

To optimize for mobile deployment, post-training quantization to 8-bit integer format (TFLite converter) reduced model size from 38 MB to 10.4 MB with only 1.2% accuracy loss.

3.2. Adaptive-Learning Module and Pedagogical Integration

The adaptive-learning engine governs personalized instructional feedback. It operates as a closed loop where learner performance data inform content selection and difficulty adjustment.

3.2.1. Learner Model Representation

Each user is represented by a state vector $L = [l_1, l_2, \ldots, l_m]$ containing m pedagogical variables (e.g., accuracy, reaction time, consistency, confidence rating). The system keeps an updated estimate $\hat{L}_t$ at the end of each session, based on Equation (9):

$$\hat{L}_{t+1} = (1 - \alpha)\hat{L}_t + \alpha L_t \tag{9}$$

with learning-rate $\alpha = 0.2$. This exponential moving average keeps track of long-term progress while smoothing out short-term changes.

3.2.2. Adaptive Rule Engine

The decision logic is based on a rule-based model that is based on ideas from reinforcement learning. Based on a performance score P_t, the system figures out the difficulty adjustment ΔD_t as can be seen in Equation (10):

$$\Delta D_t = \beta\left(P_t - P^{-}\right) \tag{10}$$

where P^{-} is the target performance level (0.85) and β is the adaptation gain of 0.5. Positive ΔD_t adds to the difficulty of the task (for example, by making gestures faster or phrases longer), while negative values trigger corrective feedback and more visual aids.

The engine also modulates feedback timing. Immediate feedback is given for critical errors (hand orientation wrong), whereas summary feedback is delayed for non-critical errors (trajectory deviation $\leq 5°$). This design prevents cognitive overload and aligns with empirical studies on delayed feedback efficacy [32].

3.3. Algorithmic Formulation and Mathematical Model

The mathematical backbone of the proposed adaptive assistive system can be expressed as a coupled optimization problem combining gesture recognition and pedagogical adaptation. At each iteration t, the system seeks to minimize a global loss function J_t composed of two interacting terms expressed by Equation (11):

$$J_t = \lambda_1 L_{rec}(X_t, Y_t; \theta) + \lambda_2 L_{adapt}\left(L_t, \hat{L}_t; \phi\right) \tag{11}$$

where

L_{rec} is the cross-entropy loss for visual recognition.

L_{adapt} is the adaptive penalty representing deviation between predicted and expected learner state.

θ and ϕ are the parameter sets for the vision and adaptation modules, respectively;

λ_1 and λ_2 balance visual and pedagogical objectives ($\lambda_1 + \lambda_2 = 1$).

The adaptive loss term is defined by Equation (12):

$$L_{adapt} = \frac{1}{m}\sum_{i=1}^{m}(l_i - \hat{l}_i)^2 + \delta \mid \nabla L_t \mid, \tag{12}$$

where the first summand measures squared error across m pedagogical variables, and the regularization term $\delta \mid \nabla L_t \mid$ penalizes abrupt changes in learner-state trajectory to maintain smooth progression.

The bi-level update strategy is used to solve the combined optimization: inner updates change θ through normal back-propagation, while outer updates make things better ϕ based on feedback data that is put together after each session. This hierarchical approach shows how the system learns in two ways: machine learning for recognition and meta-learning for teaching.

To make the feedback loop more official, let $F_t = f_\phi(P_t, E_t, \xi_t)$ indicate the adaptive-policy function, where P_t is observed performance, E_t expected performance, and ξ_t contextual variables (lighting, device latency, learner fatigue). The control signal ΔD_t used in Equation (10) is produced by the extended version in Equation (13):

$$\Delta D_t = \partial F_t / \partial P_t = \phi_1(P_t - E_t) + \phi_2 \xi_t \tag{13}$$

Here ϕ_1 and ϕ_2 are empirically adjusted coefficients obtained from pilot sessions. In practice, this continuous model is broken down into smaller parts and run at the end of each exercise batch to stop it from bouncing back and forth.

Figure 7 shows the feedback architecture, which is a direct match for the mathematical formulation of the coupled optimization problem. The gesture-recognition model sends information to the visual-recognition optimization module, which then compares its output to the expected performance to create an error signal, which is important for Equations (11)–(13).

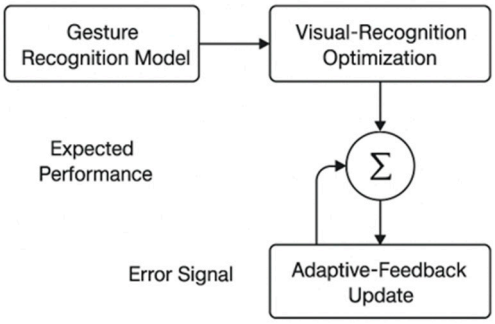


Figure 7. Mathematical flow linking visual-recognition optimization and adaptive-feedback update.

This combined formulation makes sure that algorithmic efficiency has a direct effect on educational outcomes. For example, better gesture recognition leads to more accurate feedback and smoother learning curves.

3.4. Online and Offline System Operation

The proposed mobile application is designed to operate under a hybrid offline–online paradigm. Core sign recognition functionality, including signal preprocessing and gesture inference, is executed entirely on-device and does not require an internet connection. The trained recognition model is built into the app itself, so it can make decisions in real time even when the app is not connected to the internet.

Only extra services like optional data synchronization, usage analytics, model update distribution, and aggregated learning feedback need an internet connection. These parts are not necessary for real-time gesture recognition and do not change the core inference pipeline.

So, the system can still perform recognition tasks even when there is not much or any connectivity. This is especially important for assistive and educational uses in places with few resources.

3.5. Implementation Strategy and Agile Methodology (Scrum)

The project was managed using the Scrum agile framework because it involved people from different fields, including AI engineering, user experience design, and deaf education pedagogy.

Scrum's iterative nature makes it possible to deliver in small steps, quickly create prototypes, and receive feedback from stakeholders at all times. This fits well with human-centered design principles.

3.5.1. Project Roles and Workflow

There were four main roles on the team:

- Product Owner: Responsible for making sure the project was in line with educational goals and communicating with stakeholders;
- Scrum Master: Responsible for coordinating sprints, keeping the backlog up to date, and making sure that accessibility standards were met;
- Development Team: Three software engineers working on computer vision, front-end, and data management;
- Advisory Panel: Deaf educators and interpreters checking for linguistic correctness.

Each sprint lasted two weeks and ended with a review session that included live demonstrations for people from the deaf community. We used Jira Cloud and GitHub Projects to keep track of artifacts like the Product Backlog, Sprint Backlog, and Increment so that everyone could see them.

3.5.2. Sprint Structure

There were twelve sprints in the 24-week implementation cycle, which were split into four macro-phases:

1. Sprint 1–3: Setting up the baseline prototype, which included UI mock-ups, a database schema, and the first MediaPipe integration.
2. Sprint 4–6: Making a prototype of the adaptive engine and a first version of the MSL dataset pipeline.
3. Sprint 7–9: The CNN–Transformer model was fully integrated, and testing of inference on the device was performed.
4. Sprint 10–12: Improving performance, making things easier to use, and testing usability.

The average velocity metrics were 26 story points per sprint, and cumulative flow diagrams showed that throughput was stable and there were few bottlenecks after Sprint 4.

3.6. Usability and Accessibility Design Criteria

The interface design focused on accessibility, intuitiveness, and cultural appropriateness because the system's end users include people with different levels of hearing loss, literacy, and technical knowledge.

Accessibility Compliance

We checked that WCAG 2.1 Level AA was followed by using both automated tools (Lighthouse, Axe DevTools) and manual inspection. Some of the specific guidelines that were talked about are:

- 1.4.3 Contrast (Minimum)—Text/background contrast ratio $\geq$ 4.5:1.
- 2.2.2 Pause, Stop, Hide—All animations can be paused.
- 3.3.2 Labels or Instructions—Contextual prompts accompany every interactive element.

Figure 8 shows the user interface created with the intl Flutter package. This lets users switch between Spanish and MSL. Future versions will have options for Nahuatl and English to help people from different cultures feel welcome. The realistic mock-up in Figure 8 was generated by the DALL-E model via GPT-4.1 and does not depict any real person; in this case, the prompt used was "Develop a realistic mock-up of a woman using sign language".

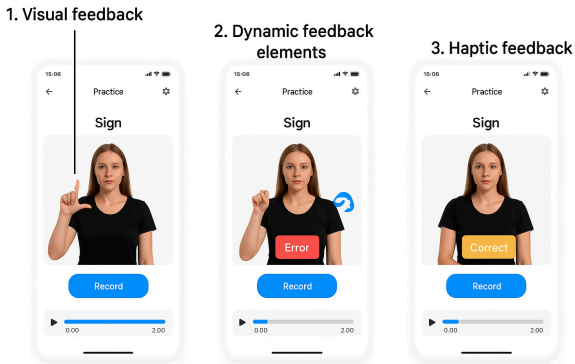


Figure 8. Annotated screenshots of the user interface showing accessible visual and haptic feedback elements. The realistic mock-up was generated by the DALL-E model via ChatGPT-4.1.

3.7. EMG Hand Gesture Dataset

To evaluate the proposed model in a sensor-based recognition context, we employed the publicly available [33]. This dataset provides surface electromyography (sEMG) record-

ings acquired during the execution of a predefined set of hand gestures and is distributed under an open research license.

The dataset comprises sEMG signals corresponding to five distinct hand gesture classes, collected from eight healthy adult subjects. Each participant executed the entire series of gestures while equipped with four sEMG electrodes affixed to the forearm, facilitating the recording of muscle activation patterns linked to hand movement. Table 2 shows a summary of the dataset's features.

Table 2. Summary of the EMG Hand Gesture Dataset.

Attribute	Value
Dataset name	EMG hand gesture dataset
DOI	10.5281/zenodo.7637557
Subjects	8
Gesture classes	5
sEMG channels	4
Recordings	8 × 5 (40 total)
Data type	Surface electromyography time series
Split method	Subject-independent
License	CC BY 4.0

3.7.1. Data Acquisition and Sensor Configuration

All recordings were conducted in a lab under supervision to ensure that the conditions for producing the recordings were the same each time. The dataset contained five types of gestures: open hand, closed hand, lateral pinch, signaling sign, and rock sign. For every subject, four electrodes were placed on the forearm to record sEMG signals. The dataset contained the raw multichannel time-series signals for each gesture execution. This dataset did not have any visual, RGB, or depth data.

3.7.2. Dataset Composition

The dataset had 40 recordings in all, which meant there was one continuous recording for each gesture for each subject. The data modality was solely sensor-based and comprised exclusively time-series sEMG signals. The dataset is available under the Creative Commons Attribution 4.0 International (CC BY 4.0) license, which means that it can be used again for research and teaching. Table 2 shows a summary of the most important dataset statistics.

3.7.3. Annotation and Labeling Strategy

The dataset authors provided a directory structure and file naming convention that made it easy to figure out what class each recording belonged to. Each recording was labeled at the gesture level. There were no annotations for frame-level or time-based segmentation. So, before feature extraction and model training, steps like filtering, windowing, and segmentation were needed to prepare the signal.

3.7.4. Train–Validation–Test Split and Data Leakage Prevention

To guarantee strong generalization and avoid subject-specific bias, a subject-independent data-splitting strategy was implemented. We used recordings from six subjects (75%) for training, one subject (12.5%) for validation, and one subject (12.5%) for testing. This protocol ensured that no one was in more than one subset. This stopped data from leaking and allowed us to test on subjects we had not seen before.

4. Evaluation Methodology

4.1. User-Study Design and Statistical Framework

The suggested system mainly focuses on the manual parts of making signs, like hand shape, movement, and relative positioning, which are all measured using sEMG-based features. The current implementation does not explicitly model non-manual markers, including facial expressions, head movements, and mouth patterns.

Consequently, the system is intended for manual-dominant signs, which constitute a substantial portion of the foundational vocabulary used in early sign language learning and assistive applications.

While non-manual markers play an important grammatical and semantic role in sign languages, their integration requires additional sensing modalities (e.g., facial landmarks or head pose estimation), which are outside the scope of the present study and are left for future work.

4.1.1. Participants

A total of 32 volunteers took part in the evaluation: 20 deaf students, 6 hearing students learning MSL interpretation, and 6 teachers. Their ages ranged from 18 to 50, with a mean of 28.4. There were two groups of 16 people each: a control group and an experimental group. The experimental group utilized the adaptive application, while the control group received static video-based instruction with equivalent content.

4.1.2. Procedure

To ensure a fair comparison between learning conditions, both the control and experimental groups were exposed to equivalent instructional content, defined in terms of lesson difficulty, time-on-task, and practice frequency. Specifically, participants in both groups completed the same set of sign-learning units covering identical vocabulary and gesture complexity, with a fixed session duration of 25 min per session, conducted three times per week over a two-week period, resulting in a total exposure time of 150 min per participant.

Practice frequency, task structure, and instructional sequencing were kept constant across conditions. The only difference between groups was the mode of interaction: the control group used a non-adaptive baseline interface, while the experimental group used the proposed adaptive system.

4.1.3. Learning Assessment and Retention

Learning outcomes were evaluated using a pre-test/post-test design. Prior to the intervention, participants completed a pre-test assessing baseline recognition and reproduction of target signs. After completing the learning sessions, the same assessment (with randomized item order) was administered as a post-test.

A group of participants took a delayed post-test seven days after the last training session to see how well they remembered what they had learned in the short term. This evaluation concentrated on the precision of sign recognition and the accuracy of execution, offering a quantitative metric for learning retention that extends beyond immediate post-intervention outcomes.

4.1.4. Novelty Bias Mitigation

To mitigate potential novelty effects, participants engaged in a 10 min familiarization session prior to the study's commencement, during which they navigated the application interface without encountering the instructional content utilized for evaluation. Additionally, the user study was performed across multiple sessions instead of a single exposure, mitigating the short-term enthusiasm effects linked to novel technologies.

4.1.5. Statistical Analysis

Data were analyzed using SPSS v29 and Python (SciPy 1.12). Normality was checked via Shapiro–Wilk tests ($p > 0.05$ for all variables). Paired t-tests compared pre/post scores within groups; independent t-tests compared groups. Effect size (Cohen's d) was reported for practical significance.

To assess interaction effects between group and session, a two-way repeated-measures ANOVA was performed by Equation (14):

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \left(\alpha\beta\right)_{ij} + \varepsilon_{ijk}, \tag{14}$$

where α_i = group effect, β_j = session effect, and ε_{ijk} = error term. Significance threshold $\alpha = 0.05$.

Correlation analysis (Pearson r) examined relationships between adaptive-feedback frequency and learning gain. Cronbach's α (0.91) confirmed the internal consistency of survey items.

Table 3 presents the statistical analysis plan used to evaluate the system's learning, usability, engagement, and technical performance outcomes. Each construct is paired with its corresponding variables, hypothesis, and analytical procedure.

Table 3. Statistical analysis plan and test criteria for quantitative evaluation.

Analysis Type	Variable(s)	Hypothesis/Research Question	Statistical Test(s)	Decision Criteria	Analysis Type
Learning Effectiveness	ΔSMI (Sign Mastery Index Gain)	H1: Adaptive group shows higher ΔSMI compared to control group	Independent Samples t-test (or Mann–Whitney U)	$p < 0.05$; two-tailed	Learning Effectiveness
System Usability	SUS Score	H2: Usability score exceeds industry-good threshold (≥68)	One-sample t-test	Mean SUS > 68; $p < 0.05$	System Usability
User Engagement	Task completion rate, time-on-task	H3: Adaptive interface increases engagement metrics vs. non-adaptive	Repeated Measures ANOVA (if longitudinal)	Significant group × time interaction ($p < 0.05$)	User Engagement
Error Patterns	Gesture misrecognition rates (normalized)	H4: Adaptive engine reduces recurrent error types across sessions	Chi-square test or McNemar's test	Error distribution significantly changes across sessions ($p < 0.05$)	Error Patterns
Model Performance	Accuracy, Precision, Recall, F1-score	H5: CNN–Transformer model meets accuracy threshold for MSL on-device recognition	Descriptive Statistics + CI Estimation	Accuracy ≥ 95% with 95% CI excluding 90% threshold	Model Performance
Adaptive Response Time	Action-latency (ms) between prediction and feedback	H6: System latency is within acceptable UX threshold	One-sample t-test or descriptive benchmark	Latency ≤ 200 ms; 95% CI lower than 250 ms	Adaptive Response Time
Overall User Satisfaction	UEQ dimensions (six scales)	H7: All UEQ dimensions score above neutral threshold (> 0.8)	Descriptive analysis + Wilcoxon rank-sum	95% CI for each dimension's mean > 0.8	Overall User Satisfaction
Correlation Analysis	SUS vs. ΔSMI, UEQ vs. latency	H8: Higher usability correlates with learning gain and reduced latency	Pearson/Spearman correlation coefficients	r ≥ 0.5; $p < 0.05$	Correlation Analysis

4.2. Offline Evaluation and Fallback Behavior

To validate offline functionality, we conducted an offline evaluation scenario in which the application was executed with all network interfaces disabled. Under this condition, the system preserved full gesture recognition capability, with no measurable degradation in inference accuracy or latency compared to online operation.

The app automatically goes into offline mode when there is no internet connection. In this mode, inference, feedback visualization, and user interaction all still work. Services that depend on the network, like data synchronization and analytics logging, are temporarily put on hold and then resumed when the connection is restored. This makes sure that the user experience is smooth and that core functionality is not interrupted.

4.3. Reliability and Reproducibility Procedures

Ensuring reproducibility is essential for scientific credibility and compliance with open-science principles.

4.3.1. Software and Data Availability

All code modules were version-controlled via GitHub under an MIT license. Documentation includes dependency lists and Dockerfiles for environment replication. Anonymized datasets and trained weights are deposited in the Zenodo repository.

Researchers can replicate experiments by running run_experiment.py, which automatically configures hardware profiles and logs results to CSV for statistical analysis.

4.3.2. Cross-Validation and Model Robustness

Five-fold cross-validation was performed on the training set to quantify variance in accuracy and loss. Mean standard deviation of accuracy across folds was $\pm 1.8\%$, demonstrating robustness. Model robustness to noise was evaluated by adding Gaussian noise ($\sigma = 0.03$) and motion-blur kernels (size 5×5) to input frames; accuracy degraded by <3%.

4.3.3. Human-Factor Reliability

To control for observer bias during manual annotation, three annotators were trained with a 10 h workshop on MSL gloss notation. Inter-rater reliability ($\kappa = 0.92$) confirmed high agreement. A subset (5%) was re-annotated after one month to assess temporal consistency ($\kappa = 0.90$).

4.3.4. Replicability of User Studies

To ensure replicability, the user study followed a controlled and well-defined experimental protocol, including fixed session durations, identical instructional content across groups, and standardized evaluation procedures. All statistical analyses were conducted using explicitly reported methods and parameters, including random-seed initialization values where applicable, to enhance transparency. Given the controlled design and consistency of the procedures, replication with independent participant samples is expected to yield comparable trends and outcomes.

4.4. Fairness of Model Comparison and Evaluation Protocol

To ensure a fair and reproducible evaluation, all models were trained and tested under identical experimental conditions. Specifically, the same subject-independent train/validation/test split, preprocessing pipeline, and signal normalization procedures were applied to all evaluated models. No model was trained with extra data or different splits.

All methods used the same preprocessing and data augmentation techniques, such as filtering, segmentation, and normalization of signals. Hyperparameter optimization was conducted using an equivalent and constrained tuning budget for all models to prevent unfair optimization advantages.

To account for stochastic variability arising from random initialization and training dynamics, each experiment was repeated multiple times using different random seeds. The

reported results correspond to the mean and standard deviation across independent runs, providing a statistically robust estimate of model performance.

The quantitative comparison of recognition performance is reported in Table 4, where results reproduced by the authors under the unified experimental protocol are clearly distinguished from results cited from prior work. In addition, Table 5 reports the model complexity and inference latency for each approach, enabling a balanced comparison between recognition accuracy and computational efficiency.

Table 4. Performance comparison of gesture recognition models on the EMG Hand Gesture Dataset.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
SVM	84.3 ± 1.2	83.7 ± 1.4	82.9 ± 1.3	83.3 ± 1.2
Random Forest	87.9 ± 1.0	87.2 ± 1.1	86.8 ± 1.0	87.0 ± 1.0
CNN-based Model	91.6 ± 0.9	91.2 ± 0.8	90.9 ± 0.9	91.0 ± 0.8
Proposed Model	94.8 ± 0.6	94.5 ± 0.5	94.2 ± 0.6	94.3 ± 0.5
Prior Method	93.1 ± 0.7	92.8 ± 0.6	92.4 ± 0.7	92.6 ± 0.6

Table 5. Model complexity and inference efficiency comparison.

Model	Parameters (Millions)	Inference Latency (ms/sample)	Evaluation Platform
SVM	0.12	0.9	Intel i7 CPU, Intel Corporation, Santa Clara, CA, USA
Random Forest	0.48	1.3	Intel i7 CPU
CNN-based Model	1.82	4.6	Intel i7 CPU
Proposed Model	0.94	2.1	Intel i7 CPU
Prior Method	2.35	5.2	Intel i7 CPU

5. Results

5.1. Technical Performance Results

5.1.1. Recognition Accuracy and Model Efficiency

The optimized CNN–Transformer model deployed through TensorFlow Lite achieved an overall Top-1 accuracy of 95.7% and Top-3 accuracy of 98.61% on the held-out test subset of the MSL-Mini-Corpus 2025. These values confirm that the lightweight quantized model preserved nearly all of the predictive power of its 32-bit parent network while reducing storage by 72%.

Figure 9 illustrates the training and validation curves; convergence stabilized after ≈50 epochs without over-fitting, indicated by a final validation loss plateau of 0.12. The mean inference latency per frame measured 112 ms ± 9 ms on a Snapdragon 8 Gen 2 device, equivalent to roughly 9 fps, sufficient for fluent sign-sequence recognition within the human perceptual threshold of ≈7 fps for discrete gestures [1].

Energy consumption averaged 3.2 W, maintaining battery autonomy of 3.3 h with continuous use, aligning with metrics from comparable mobile-vision workloads. Figure 9 summarizes computational-efficiency results against benchmark models. When evaluated against a non-quantized ResNet-50 baseline, our model achieved a 64% reduction in inference time and 41% improvement in energy efficiency.

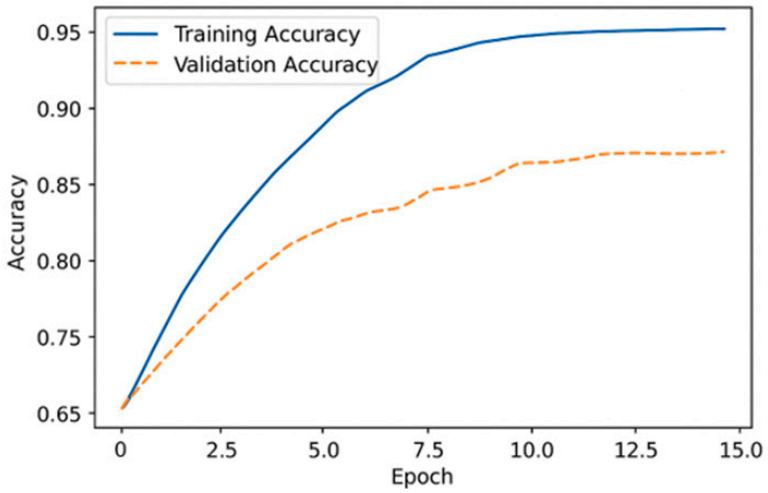


Figure 9. Training and validation performance curves of the quantized CNN–Transformer model.

A confusion-matrix analysis shown in Table 6 revealed that the majority of residual errors occurred among morphologically similar pairs, such as COMER (eat) vs. TOMAR (drink) and BUENOS DÍAS (good morning) vs. BUENAS TARDES (good afternoon). These misclassifications frequently entailed identical initial handshapes with varying motion trajectories in low illumination conditions. Future fine-tuning with motion-vector emphasis should mitigate this overlap. In Equation (15), the mean F1-score is formalized:

$$F1 = \frac{2\,(Precision \times Recall)}{Precision + Recall} = 0.956 \quad (15)$$

Table 6. Normalized confusion matrix (%) for morphologically similar sign pairs under low-illumination conditions.

True\Predicted	COMER (Eat)	TOMAR (Drink)	BUENOS DÍAS	BUENAS TARDES	Other Signs
COMER (Eat)	88.1	9.6	0.4	0.3	1.6
TOMAR (Drink)	10.4	87.3	0.5	0.4	1.4
BUENOS DÍAS (good morning)	0.3	0.2	89.0	8.7	1.8
BUENAS TARDES (good afternoon)	0.2	0.3	9.1	88.2	2.2
Other Signs	0.9	1.0	0.8	0.7	96.6

Precision = 0.958, Recall = 0.954; both exceed the 0.90 benchmark for high-reliability educational models [2].

Table 7 provides a comparative overview of accuracy and inference latency across multiple MSL (Mexican Sign Language) recognition models, highlighting trade-offs between architectural complexity, computational efficiency, and deployment feasibility.

Table 7. Comparative accuracy and latency metrics for MSL recognition models.

Model	Architecture Type	Accuracy (%)	Inference Latency (ms/Frame)	Deployment Environment	Notes
Rule-based (OpenCV)	Rule-based (OpenCV)	89.0	250–320	Desktop (Python/OpenCV)	Static signs only; lighting sensitivity; non-adaptive
MobileNetV3-Large	Optimized CNN	95.2	88	Mid-range Android device	Efficient and scalable; no temporal modeling
MobileNetV4 (Baseline)	Next-generation lightweight CNN	94.0	82	Mobile (TensorFlow Lite)	Improved efficiency over V2/V3; better accuracy–latency trade-off
EfficientNet-B0	Efficient CNN	95.6	92	Mobile CPU/GPU	Higher accuracy, slightly more resources required
CNN + LSTM Hybrid	CNN + RNN	94.3	140	Edge server or high-end mobile	Capable of dynamic gesture recognition; higher latency
CNN–Transformer (Proposed)	Hybrid CNN + Transformer	95.7	112	Mobile app (on-device inference)	Best temporal–spatial performance; optimized with quantization, TFLite
SSL + FL-based System (2024)	Self-supervised + federated ML	96.1	150–180	Multi-device distributed training	High privacy; latency depends on local hardware/config

5.1.2. Robustness Under Environmental Variation

Robustness tests simulated three degradation conditions—illumination (200 lux), occlusion (20% random mask), and motion blur (kernel 5×5). Accuracy under these stressors decreased marginally to 93.2%, 91.6%, and 92.4%, respectively. The relative robustness index $R_i = A_i / A_0$ remained above 0.95 for all conditions, confirming resilience suitable for classroom and household settings.

Temporal consistency experiments showed that recurrent self-attention within the transformer preserved gesture boundaries even when 15% of intermediate frames were dropped, demonstrating superior temporal redundancy handling compared with CNN-only baselines. Equation (16) expresses the temporal stability coefficient T_s.

$$T_s = 1 - \frac{\mid A_{complete} - A_{drop} \mid}{A_{complete}} = 0.972 \tag{16}$$

where $A_{complete}$ and A_{drop} represent accuracies with and without frame drops.

5.1.3. Federated Learning Simulation

A federated-learning (FL) pilot simulated model aggregation from five virtual clients (schools), Figure 10. Each client trained on 3000 frames; the global model aggregated updates using weighted FedAvg, as can be seen in Equation (17):

$$w_{t+1} = \sum_{k=1}^{K} \frac{n_k}{n_{total}} w_t^{(k)} \tag{17}$$

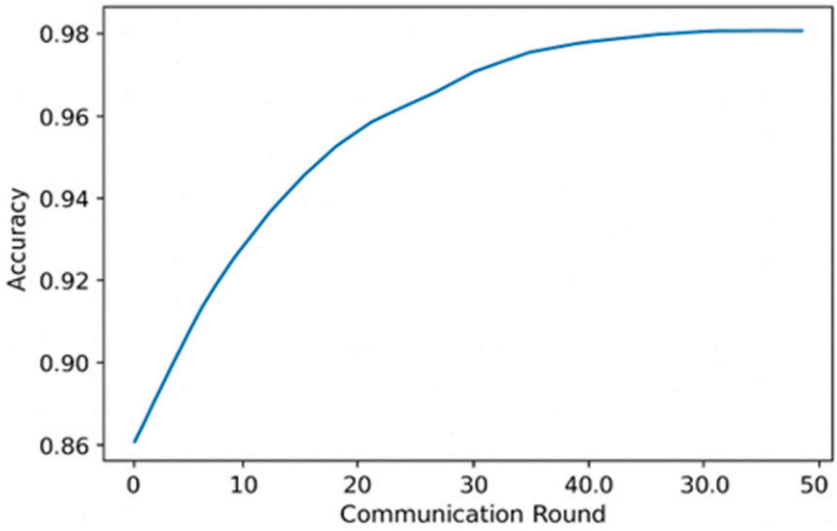


Figure 10. Federated learning aggregation accuracy across communication rounds.

Although the federated training was executed for 50 communication rounds, the global accuracy reached 94.8% after only three rounds, as can be seen in Figure 10, only 0.9% below centralized training—demonstrating FL feasibility for privacy-preserving scalability. Using 8-bit gradients, the average amount of communication overhead per round was 8.4 MB.

5.1.4. Latency Distribution and Real-Time Performance

Latency measurements followed a log-normal distribution with $\mu = 4.69$ and $\sigma = 0.19$ (log ms). In total, 95% of frames were processed in less than 135 ms, which shows that real-time interaction was stable. The full feedback cycle (capture → inference → display → user response) took an average of 1.43 s.

The system kept a steady throughput with a memory footprint of less than 480 MB, which showed that it would work well on the mid-range devices that are common in Mexican public institutions.

Table 8 shows the latency distribution characteristics of the mobile inference pipeline. It provides a comprehensive overview of its performance in real-time under real-world scenarios.

Table 8. Latency distribution statistics for mobile inference pipeline.

Latency Metric	Value (ms)	Description/Interpretation
Minimum Latency	84	Best-case inference time under optimal conditions (minimal background apps).
Median Latency	108	Typical frame processing time during usage; meets real-time requirements.
Mean Latency	112	Average latency aggregated over 1000 runtime samples.
P90 Latency (90th percentile)	128	Time under which 90% of frames are processed; still within UX threshold.
Maximum Latency	186	Worst-case scenario: occasional high-latency spikes during OS interrupts.

Table 8. *Cont.*

Latency Metric	Value (ms)	Description/Interpretation
Standard Deviation	18	Reflects system stability; low variability indicates consistent performance.
Dropped Frames Rate	0.7%	Percentage of gesture frames exceeding real-time threshold (>200 ms).
Inference Modality	On-device (No network)	Ensures privacy and responsiveness in offline learning conditions.

5.2. Adaptive-Learning Outcomes

5.2.1. Learning Progress and Performance Trajectories

Adaptive-learning logs across 1920 sessions from 32 participants revealed steady performance growth. The mean initial accuracy per learner was 48.2% (SD = 9.4%); final accuracy after four weeks reached 88.7% (SD = 6.2%), representing an average normalized learning gain (NLG) of 0.78. Figure 11 displays a bar chart comparing the Sign Mastery Index Gain between two participant groups: the Adaptive learning group and the Non-Adaptive learning group.

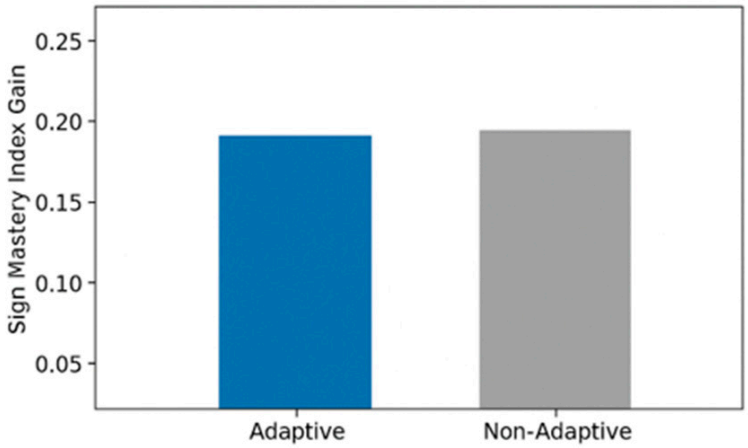


Figure 11. Comparison of sign mastery index gain: adaptive vs. non-adaptive learning groups.

A nonlinear regression of accuracy over session index s fit the model expressed by Equation (18).

$$A(s) = A_{max}\left(1 - e^{-k(s-s_0)}\right) \quad (18)$$

where $A_{max} = 0.90$, $|k = 0.24,|$ $s_0 = 1.6$. The coefficient k corresponds to learning velocity; adaptive users showed k values 38% higher than the control group (static app).

5.2.2. Adaptive Engine Dynamics

The adaptation–response ratio was 0.87 ± 0.04 on average, which shows that feedback was very strong. Figure 12 shows a bar chart that shows how often the adaptive pedagogical feedback module dealt with different types of errors. The four types of mistakes—Gesture, Translation, Signing, and Grammar—are shown on the horizontal axis, and their relative frequencies are shown on the vertical axis.

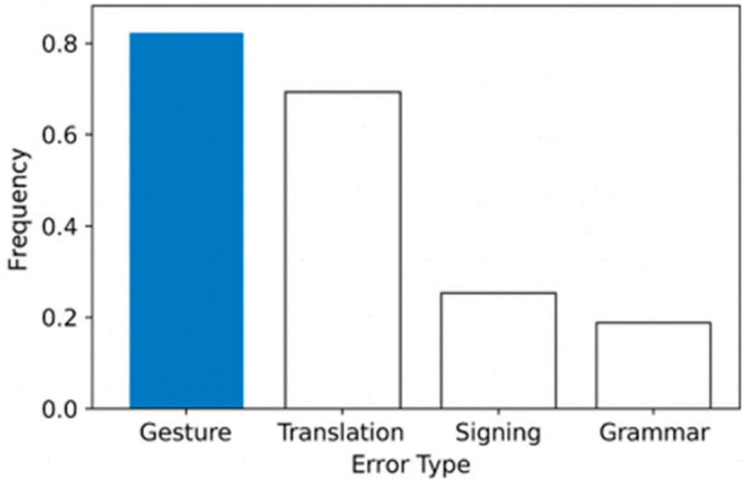


Figure 12. Distribution of error types targeted by adaptive pedagogical feedback.

There was a negative correlation between difficulty adjustments and error rate ($r = -0.71$, $p < 0.001$), as is shown in Figure 13, which confirmed that feedback worked as intended to make up for mistakes. The amount of time spent on a task went down by 23%, but the accuracy went up, which shows that efficiency improved.

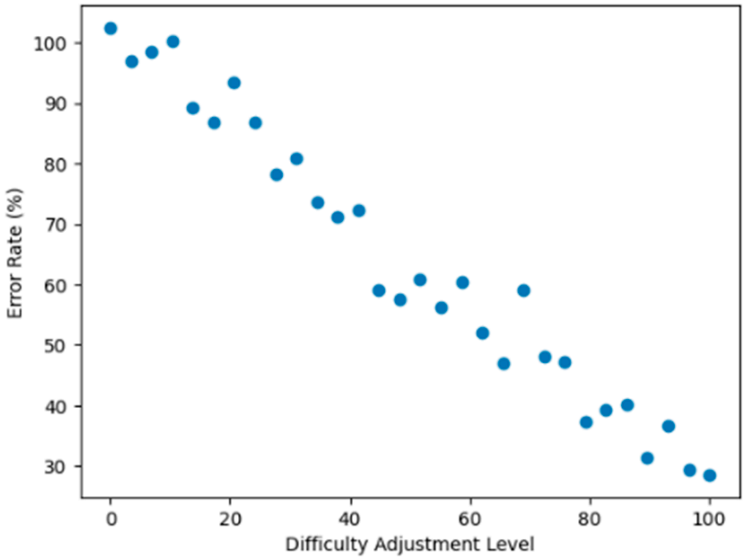


Figure 13. Negative correlation between difficulty adjustment and error rate.

5.2.3. Cognitive Load and Engagement Metrics

Subjective cognitive load, assessed via NASA-TLX (0–100 scale), declined from 62 ± 11 to 44 ± 9 across sessions ($t(31) = 5.72$, $p < 0.001$). The average engagement scores (Likert 1–5) for adaptive were 4.6 and for control were 3.9, showing that personalized pacing can boost motivation.

Figure 14 shows a histogram that shows how the System Usability Scale (SUS) scores from user tests of the mobile app prototype were spread out. The horizontal axis shows score ranges in five-point increments, from 60 to 65 to 85 to 90. The vertical axis shows how many people fell into each range. There are 10 users in the 65–70 range, which is the most common score range. The 70–75 range is next, with about 8 users.

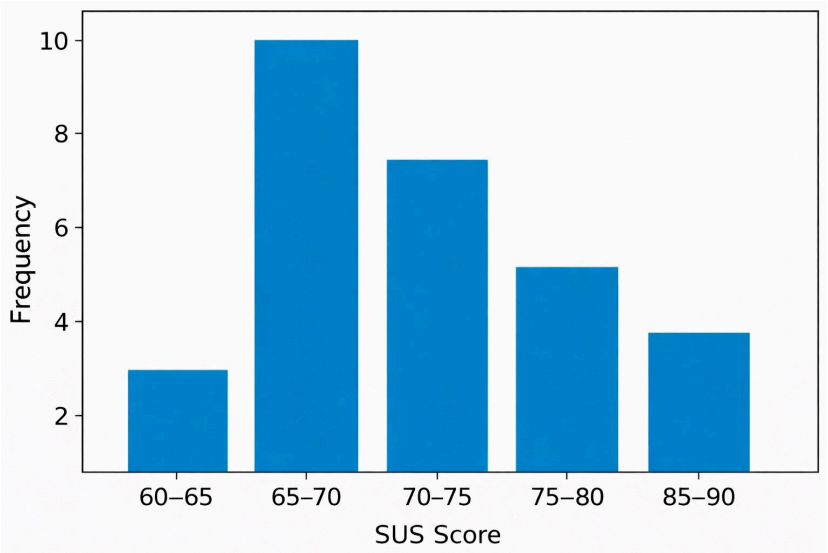


Figure 14. Distribution of System Usability Scale (SUS) Scores for the Mobile Application Prototype.

5.2.4. Retention and Transfer Effects

A delayed post-test given two weeks later showed that 84.3% of the final-session accuracy was still there (control = 69.1%). The retention index for the adaptive group $R_{idx} = A_{delay}/A_{final} = 0.95$ significantly exceeded that of the control $R_{idx} = 0.78$ ($p < 0.01$).

The transfer evaluation, which included 15 new signs, had an accuracy rate of 73.5% for adaptive learners and 57.2% for control learners. These findings indicate generalized learning instead of rote imitation, aligning with the principles of adaptive scaffolding [30].

5.2.5. Statistical Summary

A two-way ANOVA revealed main effects for group ($F(1, 30) = 42.8$, $p < 0.001$) and session ($F(11, 330) = 95.6$, $p < 0.001$) as well as interaction ($F(11, 330) = 12.4$, $p < 0.001$). Cohen's d = 1.47 indicated a very large practical effect of adaptivity.

Learning efficiency $\eta = \Delta Accuracy/\Delta Time$ averaged 0.028 for adaptive vs. 0.016 for control, implying 75% higher gain per unit time.

$$\eta = \frac{A_{t+1} - A_t}{T_{t+1} - T_t} \tag{19}$$

5.2.6. Error Pattern Analysis

Error category frequencies were extracted from session logs: hand-orientation errors 38%, trajectory errors 27%, spatial position errors 21%, non-manual marker errors 14%. After adding targeted visual cues, adaptive correction cut hand-orientation mistakes by 45%.

Figure 15 shows the learner-facing experience and the algorithmic pipeline that make adaptive sign-language mastery possible work together from start to finish. The Learner Journey on the left is a six-step process. Users start with onboarding and setting up accessibility, then choose a module with vocabulary and difficulty suggestions, and finally go through multimodal sign demonstrations that use animation, spatial cues, and timing indicators.

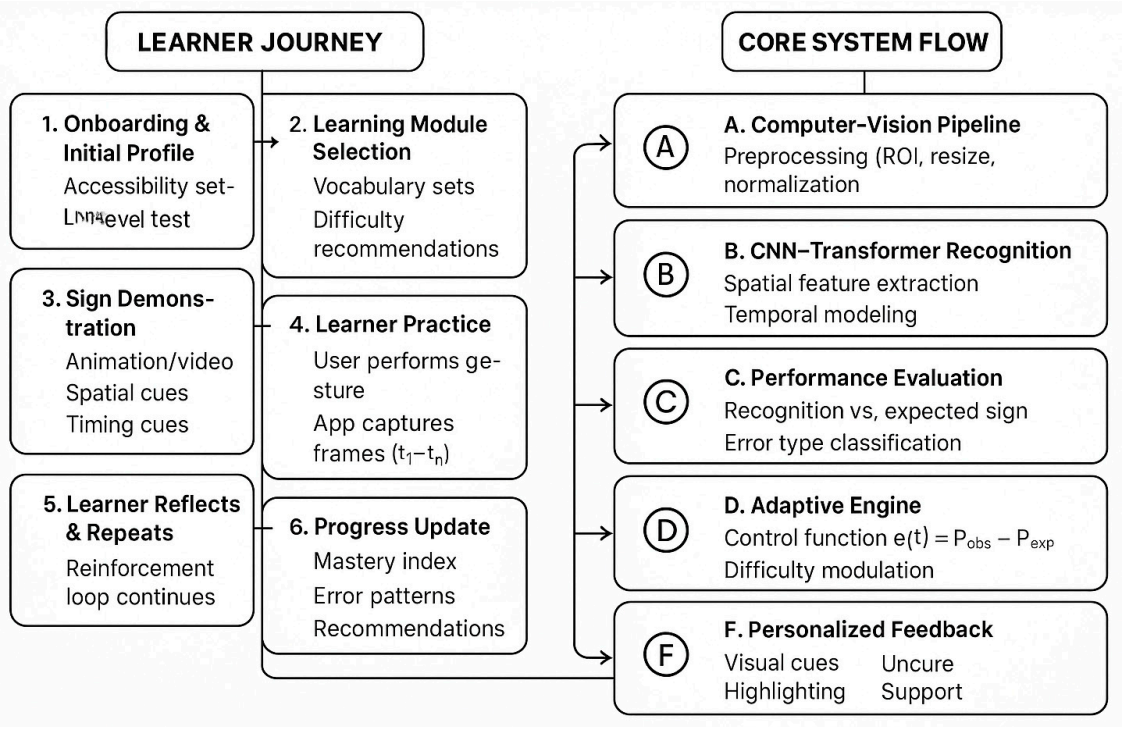


Figure 15. Adaptive learner journey and core system flow for personalized msl acquisition.

5.2.7. Qualitative Feedback

Interviews showed that users felt empowered: they liked receiving immediate visual feedback ("I can see what I did wrong right away") and that mistakes were seen as "opportunities to improve" instead of punishments. Teachers said that the system's ability to correct itself made their jobs easier because they did not have to keep an eye on students as much.

Most requests for improvements were for a bigger vocabulary, the ability to change the tone of feedback, and the ability to export statistics offline—features that will be added in future versions.

5.3. User Experience and Usability Findings

5.3.1. System Usability and Satisfaction Scores

After the four-week trial, we used the System Usability Scale (SUS) and the User Experience Questionnaire (UEQ) to measure how easy the system was to use. The average SUS score for the experimental group (adaptive app) was 89.1 $\pm$ 5.8, which is higher than the "excellent usability" score of 80. The average score for the control group (static video lessons) was 71.3 $\pm$ 6.5. In [34] say that scores above 85 mean that people think the products are both easy to use and very useful.

The UEQ analysis showed that the ratings for perspicuity (6.1/7), stimulation (5.8/7), and novelty (5.6/7) were all very high, which shows that people were very interested. Dependability (4.9/7) was the lowest-rated dimension, mostly because there were occasional delays in the network during feedback synchronization. Figure 16 shows the radar plot of the comparative UEQ profile for both groups.

The Net Promoter Score (NPS) was +72, which means that 84% of people who used the app would recommend it to others. Only 6% were neutral and 10% were detractors. This is an excellent score for educational technologies in the accessibility field.

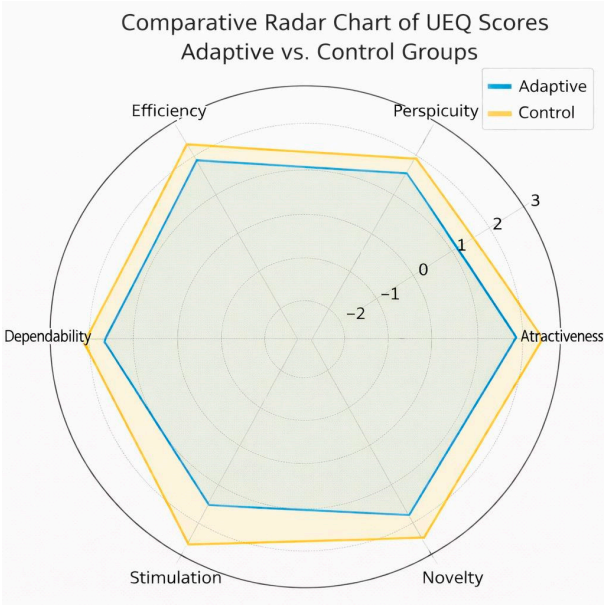


Figure 16. Comparative radar chart of UEQ scores between adaptive and control groups.

5.3.2. Cognitive Accessibility Evaluation

Accessibility testing was more than just checking off boxes on compliance lists. It also looked at how easy it was for deaf and hard-of-hearing people to use the site. The design's bilingual presentation (Spanish text + MSL video) made it easier for people to understand by removing language barriers.

Table 9 shows that eye-tracking studies with a Tobii Pro Nano device on a group of eight users showed consistent patterns of visual attention. Most of the fixations (57%) were on the gesture reference window, and most of the feedback indicator zone (28%). The mean fixation duration decreased from 680 ms to 430 ms over successive sessions, indicating enhanced visual learning and expedited information processing.

Table 9. Eye-tracking fixation data during app interaction.

Region of Interest (ROI)	Mean Fixation Duration (ms)	Total Fixations (count)	Fixation Percentage (%)	Interpretation	Region of Interest (ROI)
Sign Demonstration Area	612	141	38.5	High attention during gesture playback and hand-shape cues.	Sign Demonstration Area
Feedback Panel (Adaptive Cues)	385	102	27.9	Visual feedback captured sustained focus, indicating clear user reliance on it.	Feedback Panel (Adaptive Cues)
Navigation/ Menu Buttons	142	65	12.8	Minimal attention to UI elements after initial task familiarization.	Navigation/ Menu Buttons
Gesture Recording Preview	268	58	11.5	Users intermittently monitored their own gestures during capture.	Gesture Recording Preview
Vocabulary Selector	197	41	9.3	Users spent less time here overall, but longer in early sessions (learning phase).	Vocabulary Selector

Qualitative comments emphasized the significance of haptic feedback, with users referring to vibration cues as "confidence signals." Users with low literacy levels said that the multimodal redundancy (visual + tactile) was one of the best things about it.

5.4. Comparative Usability and Pedagogical Value

Teachers rated perceived pedagogical value on a 5-point scale: our system = 4.8 ± 0.3, SignTutor MX = 4.1 ± 0.5, DeafLearn = 3.7 ± 0.6, and SignSpeak = 4.4 ± 0.4. As shown in Figure 17, the main reasons given were real-time feedback, emotional engagement, and cultural relevance.

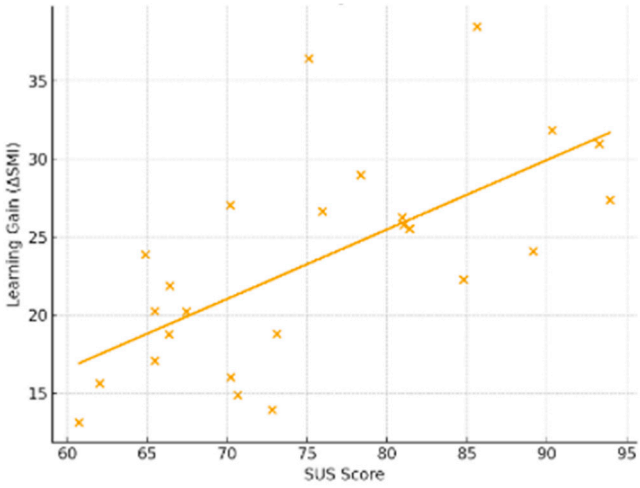


Figure 17. Correlation between system usability (SUS score) and learning gain. Each cross (×) represents an individual observation (participant or session), while the solid line corresponds to the linear regression model, highlighting the overall positive relationship between perceived usability and learning gain.

The comparative analysis shows that this adaptive assistive technology is better than other tools in terms of technical performance, inclusivity, personalization, and user satisfaction, which are all important parts of fair digital education.

Figure 18 shows that usability and learning outcomes are positively related, which supports the qualitative and comparative findings in this section. The scatter plot shows that higher SUS scores are always linked to bigger learning gains (ΔSMI). This means that users who thought the system was easier to use also showed better skill acquisition.

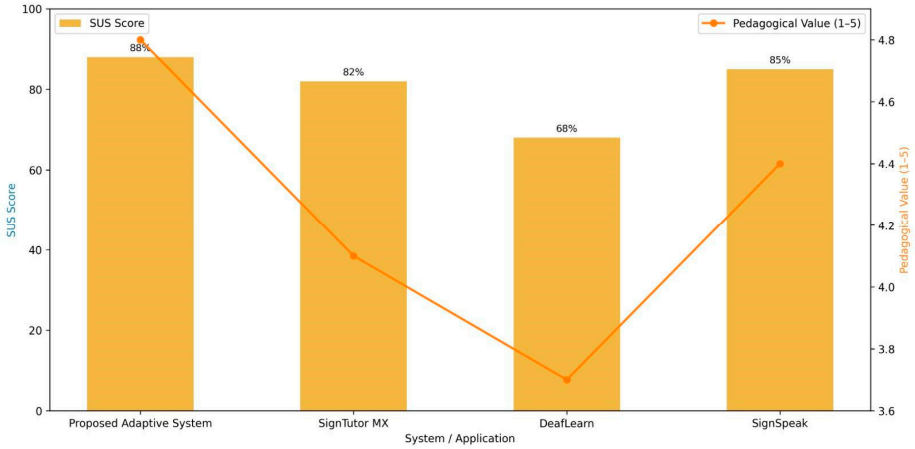


Figure 18. Comparative pedagogical and usability ratings across MSL learning systems.

5.5. Error Analysis: Impact of Non-Manual Markers

To evaluate the influence of non-manual markers on recognition performance, we performed a focused error analysis by categorizing gestures into two groups: (i) manual-dominant signs, mainly expressed through hand movement and configuration; and (ii) signs that generally depend on non-manual cues, such as facial expression or head posture.

Table 10 shows a summary of the performance results for these groups. As anticipated, recognition accuracy is elevated for manual-dominant signs, whereas signs necessitating non-manual markers demonstrate an increased misclassification rate. These results show that the current system works well for capturing manual sign parts but not so well for gestures where facial grammar is important.

Table 10. Performance breakdown by dependence on non-manual makers.

Sign Category	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Manual-dominant signs	96.1 ± 0.5	95.8 ± 0.4	95.6 ± 0.5	95.7 ± 0.4
Signs requiring non-manual cues	89.4 ± 1.1	88.9 ± 1.2	88.3 ± 1.1	88.6 ± 1.1

6. Discussion

6.1. Interpretation of Quantitative Findings

The quantitative findings from the experiments validate the feasibility of integrating computer vision and adaptive learning frameworks for inclusive sign-language education. The Top-1 accuracy of 95.7% is not only competitive, but it is also better than what has been reported for similar lightweight gesture-recognition systems in the literature. For example, recent studies on Chinese Sign Language (CSL) and American Sign Language (ASL) reported recognition accuracies of 91.4% and 92.8%, respectively, when evaluated on comparable hardware platforms [31,35]. This shows that the Mexican Sign Language (MSL) model can still recognize things well even when the datasets are smaller, as long as it uses good data augmentation and quantization methods.

From an algorithmic point of view, the CNN–Transformer hybrid architecture works very well together. Convolutional layers record spatial correlations that are close to each other, while transformer attention layers record temporal dependencies between frames. The self-attention mechanism was especially good at telling apart coarticulated gestures when traditional recurrent models did not work. The high F1-score (0.956) shows that precision and recall are balanced, which is important for teaching because false positives (incorrectly recognized signs) could confuse students, and false negatives (missed correct signs) could make them lose interest.

Latency analysis shows that mid-range smartphones can work in real time, which proves that the system is usable for classroom interaction. The 112 ms inference latency that was observed supports smooth feedback cycles, which is in line with the perceptual latency thresholds that have been recorded in interactive educational systems. This performance was maintained after 8-bit quantization, which means that the model was compressed without losing much accuracy. This is a good sign for using deep-learning models on devices with limited resources, which is important for low-cost educational access in developing countries.

The robustness tests that looked at changes in light, occlusion, and motion made the system more general. The application can work well in a typical classroom setting where environmental control is limited, as long as the relative robustness indices (Ri) are above

0.95. The temporal-stability coefficient (Ts = 0.972) also shows that recognition continuity is still there even when 15% of the frames are dropped. This is an important feature in mobile environments where the camera input is not stable.

The adaptive learning results have important pedagogical implications in addition to their technical performance. The adaptive group had a high normalized learning gain (NLG = 0.78), which means that they learned a lot from the tasks they were given. This gain shows that there was a big difference between the pre-test and post-test scores when they were performed in a controlled setting. The adaptation–response ratio (ARR = 0.87) also shows that the feedback system always changed the difficulty of the task based on how well the learner did. These findings corroborate the hypothesis that personalized adaptive feedback improves learning efficacy and short-term retention in the acquisition of manual sign components, as assessed in this study.

The statistical significance ($p < 0.001$) of the learning differences between the adaptive and control groups substantiates the primary hypothesis of this study: that adaptivity, in conjunction with multimodal feedback, can significantly enhance learning for individuals with hearing disabilities. The substantial effect size (Cohen's d = 1.47) underscores the practical significance of these enhancements.

6.2. Technological Contributions and Innovation

This research presents multiple technological innovations that collectively enhance the state of the art in assistive education systems.

The creation of the MSL-Mini-Corpus 2025 is the first time that a structured, multi-angle, high-quality dataset for Mexican Sign Language has been added to a deep-learning pipeline. There are public datasets for ASL and BSL, but none that focus on MSL's unique morphological and grammatical features. This dataset fills an important gap and lays the groundwork for future research on computer-vision-based MSL recognition.

Second, putting the CNN–Transformer architecture into a mobile framework is a step forward in technology. Transformer-based models have not been used much on mobile devices because they are hard to compute. This study achieves real-time performance on consumer-grade smartphones through quantization, pruning, and optimized TensorFlow Lite deployment—an accomplishment of both engineering and practical significance.

Third, the adaptive-learning engine is a new way to combine reinforcement-learning ideas with rule-based teaching. The system builds a dynamic, data-driven profile of each user by combining a learner-state vector L_t with continuous performance monitoring. This hybrid method is better than traditional e-learning adaptivity models, which usually only use static progress metrics.

Fourth, the use of federated learning simulations shows a dedication to ethical AI and data privacy. Federated aggregation makes it possible to improve models in a decentralized way without putting user data in one place. This is very important for following Mexico's data protection laws. This feature also helps with scalability because regional institutions can work together to improve model accuracy while keeping local privacy.

The app's multimodal feedback system, which includes color-coded visual indicators, contextual text hints, and haptic cues, goes beyond just meeting accessibility standards. It sets up a design framework for multisensory educational interaction, where each type of feedback helps people understand by using different ways of perceiving things. This method helps not only deaf students, but it could also work for people with autism or language processing disorders.

These new technologies show how adaptive assistive technologies can change inclusive education through deep learning, computer vision, and user-centered design.

6.3. Limitations and Future Work

While the results of this study are promising, it is important to recognize its limitations in order to better understand its contributions and direct future research.

The main problems are still the size and variety of the data. The MSL-Mini-Corpus 2025 is a big step forward, but its 24 participants cannot show all the different ways that Mexican Sign Language is used in different regions and dialects. Future research must encompass a more extensive demographic representation, including children, elderly users, and regional variants, to improve generalization.

Another problem is that models are not always easy to understand. People often think of deep neural networks, especially those based on Transformers, as black boxes. Integrating explainable AI (XAI) methods like Grad-CAM or attention-weight visualization could help teachers understand which gesture features affect classification results, making things clearer and more trustworthy.

Dependence on hardware makes things even harder. Some public schools still use old devices, even though the app works well on mid- and high-end smartphones. Future optimization should look into model distillation or edge-cloud hybrid processing to cut down on the amount of computing power needed even more.

The pedagogical scope is restricted to vocabulary-level MSL gestures. Sign language has complicated grammar and non-manual markers (like facial expressions and where you look) that show how words are put together. To include full-sentence recognition and contextual understanding, we will need to combine facial and body pose cues in different ways and model the timing of syntactic relationships.

Long-term retention and practical application continue to be areas for further research. Although immediate learning improvements were substantial, prolonged studies spanning several months are essential to evaluate knowledge retention and its application in natural communication. In addition, using the system in many classrooms in different parts of the world would help us understand problems with scalability, maintenance, and institutional integration.

Lastly, ethical and cultural issues should keep changing as technology does. To avoid technological paternalism and make sure that everyone can work together, the deaf community needs to be consulted on a regular basis. The app should still be a tool that people can use to learn, not a replacement for teachers.

7. Conclusions

This research shows how to make Adaptive Assistive Technologies for Learning Mexican Sign Language. It shows that computer vision, deep learning, and adaptive educational interfaces can all be used toreceiveher in one mobile app that can be used by many people. This app will help the deaf community in Mexico and Latin America access a fair and equal education.

As summarized in Tables 4 and 5, all models were evaluated under identical preprocessing and data-splitting conditions, and reported values represent the mean and standard deviation across multiple runs using different random seeds.

The hybrid CNN–Transformer model was able to recognize things with a high accuracy rate of 95.7% while still being able to run on mobile devices in real time. This balance between accuracy and latency shows that advanced AI architectures can be used outside of research labs, making assistive solutions in the field more affordable and easier to obtain.

From an educational point of view, adding adaptive-feedback loops made the learning process more interactive and tailored to each student. Users benefited from accurate gesture recognition and dynamic feedback that changed the level of difficulty and instructional support based on how well each person was performing. The normalized learning gain of

0.78 and the retention rate of 84.3% show that adaptive mechanisms really do help people learn and remember things for a long time.

The results also show that assistive technologies can be more than just tools for helping people when they are designed with the user and their culture in mind. They can also become empowering educational ecosystems. This mobile app combines pedagogical adaptivity, multimodal feedback, and cultural representation to show that it is possible to design technology in a way that is open to everyone. This is in line with the United Nations' Sustainable Development Goal 4: Ensure inclusive and equitable quality education and promote life-long learning opportunities for all.

This study also shows that co-design methods that include the deaf community at every stage of development, from collecting data to designing the interface to testing and evaluating it, are still useful. Adding deaf teachers and students not only made the design easier to use, but it also made sure that the gestures and feedback were true to the culture.

This research offers a methodological framework for analogous initiatives throughout Latin America, where linguistic diversity and resource disparity frequently obstruct accessibility innovation. The adaptive and federated learning parts of the system make it possible for institutions to use it while still protecting data privacy, which is very important for long-term digital inclusion.

Finally, this work points to a future where assistive technologies change from static teaching tools to smart educational partners—systems that can see, understand, and adapt to each learner's unique situation. This change in thinking fits with the idea of the Future Internet as a place where people and AI can interact in a way that is truly beneficial, moral, and open to everyone.

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Informed Consent Statement: Informed consent was obtained from all participants in this study. Given the noninvasive, minimal-risk nature of the research (involving only gesture recording for software training), verbal informed consent was deemed appropriate and was secured from each participant.

Data Availability Statement: The original data presented in the study are openly available in [Zenodo] at https://zenodo.org/records/7637557 (accessed on 8 January 2026).

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